

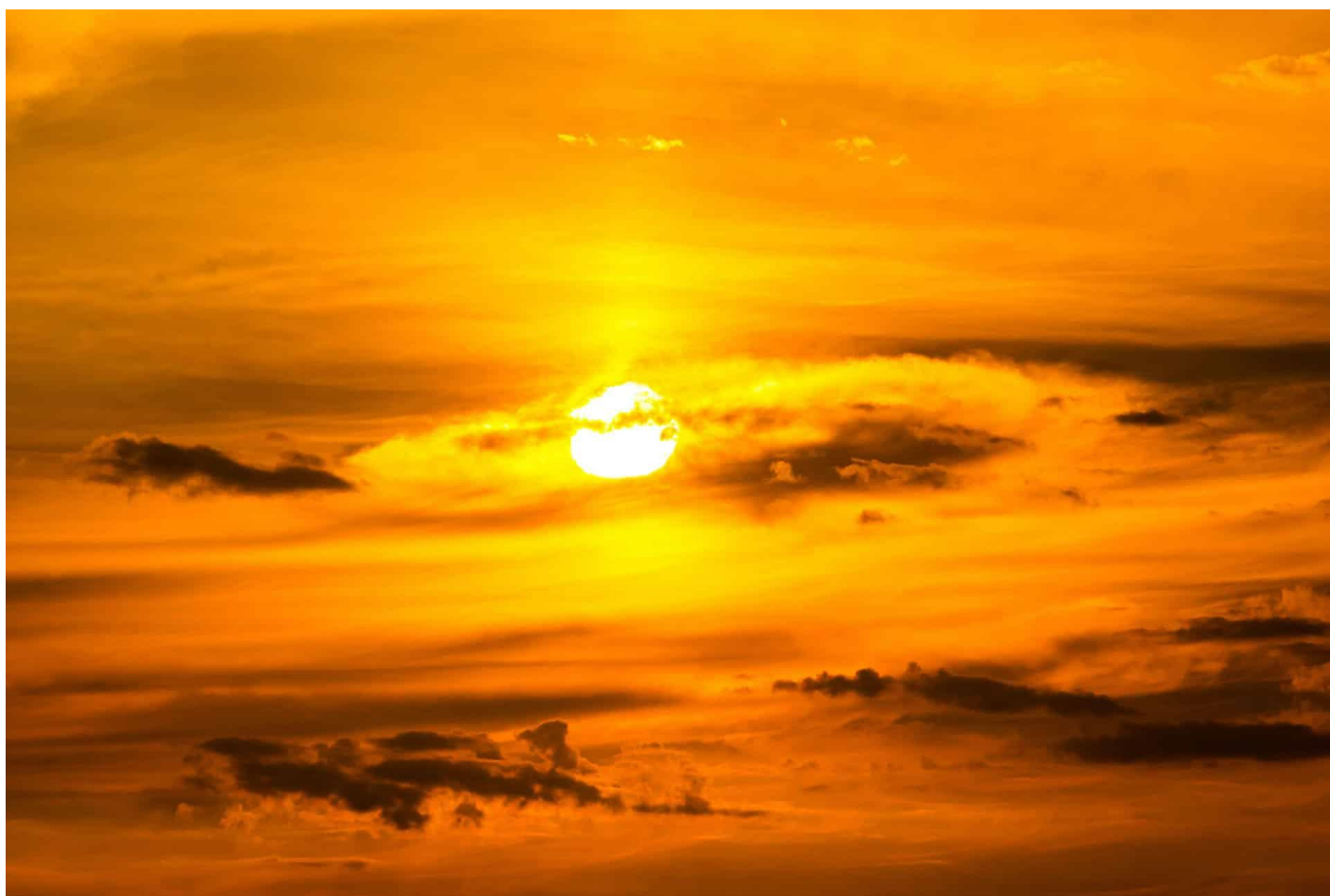


UNIVERSITY OF
ILLINOIS CHICAGO

DEPARTMENT OF CHEMICAL ENGINEERING

TRANSPORT PHENOMENA II

CHE 312, SPRING 2026



Department of Chemical Engineering
College of Engineering
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TRANSPORT PHENOMENA II

CHE 312, SPRING 2026

I. INSTRUCTOR & COURSE DETAILS

INSTRUCTOR

KOROSH TORABI

Email address: korosh@uic.edu

Drop-In Office Hours: Mondays 1:00 PM -2:00 PM, EIB 246

TEACHING ASSISTANT

KIM NGO

E-mail: bngo@uic.edu

Drop-In Office Hours: Wednesdays 2:00 PM – 3:00 PM, EIB 247

COURSE BLACKBOARD SITE

<https://uic.blackboard.com>

Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. Email the Learning Technology Solutions team at LTS@uic.edu for all technical questions about Blackboard.

COURSE MODALITY AND SCHEDULE

This course is taught ON CAMPUS.

Tuesdays and Thursdays 1:00 PM - 2:15 PM, EIB Room 124

II. COURSE INFORMATION

CATALOG COURSE DESCRIPTION AND PREREQUISITE/COREQUISITE STATEMENT

Heat and mass transport phenomena. Heat conduction, convection, and radiation. Heat exchanger design.

Diffusion. Mass transfer coefficients.

Prerequisite(s): CHE 311.

BRIEF LIST OF TOPICS TO BE COVERED

Fourier's law of heat conduction. Fick's law of mass diffusion. The energy flux and mass flux. Application of shell balances and equations of change in energy and mass transfer problems. Turbulent temperature and concentration characteristics. Interphase transport in non-isothermal and multicomponent systems. Heat exchangers. Diffusion inside a porous catalyst. Energy and mass balance for macroscopic systems.

COURSE RATIONALE AND GROWTH MINDSET

This course builds on the material covered in CHE 201, 210, 301, and 311, with an emphasis on non-isothermal mixtures involving energy and mass transport. An ability to perform energy and mass balances will be essential in subsequent undergraduate chemical engineering courses, including Unit Operations Laboratory, Design, and Control.

This "engineering science" course combines mathematics and physics in applications for chemical engineers. Students are expected to have a strong foundation in elementary mathematics, which will be utilized throughout the course. Some review of mathematics may be necessary at various points.

It is strongly suggested that you read the textbook carefully, do the homework problems in detail without consulting other solutions, ask questions in class and during office hours, discuss the material with your classmates, and work on additional problems. Course materials and assignments can be complex and challenging, but crucial to intellectual and personal growth and development. There are times when you may need extra help. Students who attend lectures, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, attend office hours, and contact faculty when struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

COURSES LEARNING OUTCOMES

By the end of this course, students should learn to:

- Develop a clear understanding of the basic mechanisms of heat and mass transfer in physical processes encountered in daily living experiences and chemical engineering practice.
- Develop the skills to describe heat and mass processes using mathematical models in laminar and turbulent systems.
- Make appropriate assumptions to simplify the equations of change for non-isothermal and multicomponent systems.
- Perform derivations and calculations to obtain qualitative and quantitative information, leading to better design and understanding of heat and mass transport processes.
- Define and apply several dimensionless quantities related to transport phenomena.
- Apply the conservation statement for energy and mass to macroscopic systems and unit operations.

ABET STUDENT LEARNING OUTCOMES

- ABET Student Outcome 1: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- ABET Student Outcome 7: an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

REQUIRED TEXTBOOK

Transport Phenomena, Revised 2nd Edition

R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot
John Wiley & Sons, Inc.; 2007

III. COURSE POLICIES & CLASSROOM EXPECTATIONS

GRADING POLICY AND POINT BREAKDOWN

- Quizzes: 20%
- Homework: 10%
- Midterm Exam I: 20%
- Midterm Exam II: 20%
- Comprehensive Final Exam: 30%

ASSIGNMENTS

- **Reading assignments:** Assigned from the textbook. No submission is required.
- **Problem sets (Homework):** Please submit an electronic copy of your homework by the specified due date before the start of the class.

POLICY FOR MISSED OR LATE WORK

- **Homework:** Late submissions will not be considered for partial credit. Unreadable and unorganized homework will not be graded.
- **Quizzes:** No makeup will be administered for the missed quizzes. However, one quiz with the lowest grade will not be counted towards your final grade.
- **Exams:** No makeup will be provided for missed exams. Students who need to miss an exam for any reason are expected to contact the instructor, preferably before the exam date. With the appropriate documentation, the grade for the missed exam will be calculated based on the average of the other exam grades.

OTHER COURSE POLICIES

Regarding exams, quizzes, and homework, requests must be submitted within two days of the grades being returned. For homework issues, please contact the TA.

ATTENDANCE AND PARTICIPATION POLICY

Please email me if you face an unexpected situation that may impede your attendance, participation in required class and exam sessions, or timely completion of assignments.

EMAIL EXPECTATIONS

Students are responsible for all information instructors send to their UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

COURSE BLACKBOARD SITE

<https://uic.blackboard.com>

Students are expected to log in to the course site regularly to stay informed about course developments, upload assignments, and communicate with classmates. Email the Learning Technology Solutions team at LTS@uic.edu for all technical questions about Blackboard.

ACADEMIC INTEGRITY

As a student and member of the UIC community, you are expected to adhere to the [Community Standards of academic integrity](#), accountability, and respect. Please review the [UIC Student Disciplinary Policy](#) for additional information.

IV. COURSE SCHEDULE

Week	Topics	Book Chapter
1	Heat flux and Fourier's law of heat conduction	9
1	Convective transport of energy and flow work	9
2	Shell energy balance and heat conduction in solids	10
3-4	Shell energy balance and energy transport in fluids	10
5	Energy equations of change	11
6	Energy transport problems with ordinary differential equations	11
7	Energy transport problems with partial differential equations	12
8	Midterm Exam 1	
8	Fick's law of binary diffusion	17
8	Mass and molar transport by convection	17
9	Diffusion problems with shell balance	18
10	Diffusion and reaction problems	18
10	Equations of Change for multi-component systems	19
11	(March 23-27) Spring break. No classes.	
12	Midterm Exam 2	
12	Turbulent Temperature and Concentration Characteristics	13 and 21
12	Interphase transport in non-isothermal and multi-component systems	14 and 22
13-16	Macroscopic Energy Balances and Heat exchanger design and analysis	15
17	Final Exam	Comprehensive

- Quizzes will be scheduled during regular class time and location with at least two days' notice.
- Midterm exams will take place at the regular class time and location.
- The final exam will be held on the date and time specified by the university's Registrar's Office.

DISCLAIMER

This syllabus is intended to provide students with guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as needed to address course requirements. I will communicate such changes in advance through in-class announcements and in writing via Blackboard Announcements.

V. ACCOMMODATIONS

DISABILITY ACCOMMODATION PROCEDURES

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc@uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodation, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

RELIGIOUS ACCOMMODATIONS

Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: "YOUR NAME: Requesting Religious Accommodation."

VI. CLASSROOM ENVIRONMENT

INCLUSIVE COMMUNITY

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

NAME AND PRONOUN USE

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are he/him. I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

COMMUNITY AGREEMENT/CLASSROOM CONDUCT POLICY

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to "think out loud," learn something new, and understand the concepts or experiences that guide other people's thinking.
- Debate concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.

CONTENT NOTICES AND TRIGGER WARNINGS

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

VII. RESOURCES: ACADEMIC SUCCESS, WELLNESS, AND SAFETY

We all need the help and the support of our UIC community. Please visit my office hours for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

ACADEMIC SUCCESS

- UIC [Tutoring Resources](#)
- College of Engineering [tutoring program](#)
- [Equity and Inclusion in Engineering Program](#)
- [UIC Library](#) and [UIC Library Research Guides](#)
- [Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.
- [Student Guide for Information Technology](#)

WELLNESS

- Counseling Services: You may seek free and confidential services from the Counseling Center at <https://counseling.uic.edu/>.
- Access [U&I Care Program](#) for assistance with personal hardships.
- Campus Advocacy Network: Under Title IX, you have the right to an education free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at <http://can.uic.edu/>.

SAFETY

- [UIC Safe App](#)—PLEASE DOWNLOAD FOR YOUR SAFETY!
- [UIC Safety Tips and Resources](#)
- [Night Ride](#)
- [Emergency Communications](#): By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.